New and emerging treatments for anxiety disorders

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Series explanation: State of the Art Reviews are commissioned on the basis of their relevance to academics and specialists in the US and internationally. For this reason they are written predominantly by US authors.

ABSTRACT

Anxiety disorders affect a large portion of the population. While standard treatments show efficacy, rates of treatment non-response and relapse highlight the need to augment existing treatments or develop new ones. This review summarizes new and emerging stand-alone treatments and treatment augmentation strategies for anxiety disorders. Novel stand-alone treatments include newer psychotherapies (reward-based treatments and safety behavior reduction strategies); an interoception enhancement technique (floatation-REST); transcranial magnetic stimulation (TMS) or transcranial direct current stimulation (tDCS); pharmacological agents (maritupirdine); and natural supplements—including ashwagandha and L-theanine. Novel treatment augmentation strategies all aim to improve exposure therapy, which include new behavioral techniques (inhibitory retrieval model of exposure therapy, behavioral experiments targeting intolerance of uncertainty) or pharmacological agents (*D*-cycloserine, scopolamine). Cannabidiol has been studied as a stand-alone treatment and augmentation strategy. Most of these treatments have shown at least some efficacy. Some novel treatments can be implemented clinically, while others await further testing.

Introduction

Anxiety disorders are characterized by excessive and enduring fear, anxiety, or avoidance of perceived dangers or threats that are either external (such as social criticism) or internal (such as aversive interoceptive sensations). Avoidance ranges from outright refusal to approach or premature escape from certain situations or activities, to subtle reliance on others or certain behaviors or objects in order to cope. These symptoms are associated with considerable distress or impairment in functioning and last typically six months or more in adults. The major anxiety disorders classified within DSM-5-TR[1] and ICD-11[2] include separation anxiety disorder, selective mutism, specific phobia, social anxiety disorder, panic disorder, agoraphobia, and generalized anxiety disorder (see box 1 for full list). Panic attacks, or abrupt rushes of fear including at least four out of 13 physical symptoms (such as heart palpitations, lightheadedness) and cognitive symptoms (such as fear of dying) can accompany any of the anxiety disorders as a specifier and are the focus of concern with panic disorder.

Most anxiety disorders start during childhood, adolescence, or early adulthood.[3] While they tend to be chronic if untreated,[4] substantial symptom reduction can occur,[5] although relapse is common.[6] Anxiety disorders are almost twice as common in women than men, with sex differences emerging throughout adolescence.[7] Anxiety disorders often precede other mental health conditions, including depression and substance use,[8] or co-occur with depression (nearly half of all cases),[9] rendering them a risk for disease burden. Occurrence of one or more anxiety disorders in the same individual is common.[10] They also co-occur with medical diseases and possibly contribute to them, including cardiovascular, gastrointestinal, or pulmonary disease; chronic pain; and migraine headache.[11]

This review discusses the evidence about new and emerging treatments for anxiety disorders and their clinical implementation and gaps where further research is needed. Whereas reviews often focus on one or more established treatment(s), this review focuses only on novel treatments or novel augmentations of established treatments. Our intended audience is clinicians and scientific professionals working in clinical or research settings, regardless of specific expertise on anxiety disorders.

Sources and selection criteria

To identify new and emerging treatments for anxiety disorders, we searched PubMed and EMBASE for articles between 2019 and 2024. Key terms were titles that included “anxiety” or “fear” and titles/abstracts that included “novel,” “emerging,” “innovative,” “optimizing,” “new,” “nascent,” or “incipient.” Box 2 lists the inclusion and exclusion criteria. We additionally identified novel stand-alone treatments and treatment augmentation strategies outside of this systematic database search and include them in this review. These articles were identified due to the authors’ pre-existing knowledge of the literature of

Box 1: Anxiety disorders*

- *Separation anxiety disorder*—Developmentally inappropriate and excessive fear or anxiety concerning separation from those to whom the individual is attached.
- *Selective mutism*—Consistent failure to speak in specific social situations in which there is an expectation for speaking despite speaking in other situations.
- *Specific phobia*—Marked and persistent fear of a specific object or situation:
  - Animal: specific animals or insects
  - Natural environment: heights, storms, water, etc
  - Blood-injection-injury: needles, invasive medical procedures, etc
  - Situational: airplanes, elevators, enclosed places, etc
  - Other: choking, vomiting, costumed characters, etc
- *Panic disorder*—Recurrent, unexpected panic attacks (ie, a surge of intense fear or discomfort that peaks within minutes) and persistent concern or worry or maladaptive behavioral change due to the panic attacks
- *Agoraphobia*—Marked fear or anxiety of two or more situations in which escape might be difficult or help might not be available in the event of panic-like, incapacitating, or embarrassing symptoms
- *Social anxiety disorder*—Marked and persistent fear or anxiety that occurs in one or more social situations, such as social interactions, doing something while being observed, or performing in front of others. Individual is concerned they will be negatively evaluated by others
- *Generalised anxiety disorder (generalized anxiety disorder)*—Excessive anxiety and worry occurring most days for six or more months about a number of events or activities; difficulty controlling the worry and physical symptoms
- *[Substance]-Induced Anxiety Disorder (Substance/Medication-Induced Anxiety Disorder)*—Panic attacks or anxiety are prominent, developed during or soon after substance or medication use, and the substance or medication is capable of producing the panic attacks or anxiety. Not better explained by another anxiety disorder
- *Secondary anxiety syndrome (anxiety disorder due to another medical condition)*—Panic attacks or anxiety are prominent, due to a medical condition, and not better explained by another anxiety disorder
- *Other specified anxiety or fear-related disorders (other specified anxiety disorder)*—Symptoms of an anxiety disorder cause distress or impairment but do not meet full criteria for other anxiety disorders; clinician specifies which anxiety disorder the presentation does not meet criteria for
- *Anxiety or fear-related disorders, unspecified (unspecified anxiety disorder)*—Symptoms of an anxiety disorder cause distress or impairment but do not meet full criteria for other anxiety disorders; clinician does not specify which anxiety disorder the presentation does not meet criteria for

*Disorder name based on ICD-11 (and DSM-5-TR, if different)

anxiety disorder treatment. Fig 1 shows the flow of selected articles; fig 2 shows the novel interventions.

Epidemiology

Together, the anxiety disorders constitute the largest group of mental disorders worldwide and are a leading cause of disability.[12 13] They are also the most prevalent mental disorder among children and adolescents.[14] Data from the 2019 Global Burden of Disease study[13] show that the global incidence of anxiety disorders increased by 47.19%, from 31.13 million in 1990 to 45.82 million in 2019. During the covid-19 pandemic, impact indicators (infection rates and reductions in mobility) were associated with increased prevalence of anxiety disorders, especially for females and younger age groups; overall, the pandemic was estimated to result in a 25.6% increase for total prevalence of anxiety disorders in 2020.[15] Rather than racial or ethnic groups, anxiety disorder prevalence and associated mental health outcomes may be better understood in relation to social determinants, with evidence for anxiety to associate with income, education, housing and food security, neighborhood safety, discrimination, and access to resources.[16]

Box 2: Inclusion and exclusion criteria of data search

Inclusion criteria

- Written in English
- Published between 2019 and 2024
- Novel interventions
- Novel augmentation strategies for well-established interventions
- Primary focus on anxiety disorders or clinically severe anxiety (e.g., using established questionnaire-based clinical severity cutoffs)
- Treatment outcome is anxiety disorder symptoms (not state anxiety)
- Randomised controlled trial
- Repeated treatment sessions or doses

Exclusion criteria

- Well established interventions without novel augmentation strategies
- Duplicate articles

Current treatments

Current first-line treatments for anxiety disorders include psychotherapy (mostly cognitive behavioral therapy (CBT)) and pharmacotherapy, with some variations across the different anxiety disorders. In terms of pharmacotherapy, selective serotonin reuptake inhibitors (SSRIs) and serotonin and norepinephrine reuptake inhibitors (SNRIs) are considered efficacious across most anxiety disorders (the meta-analyses reported were focused on social

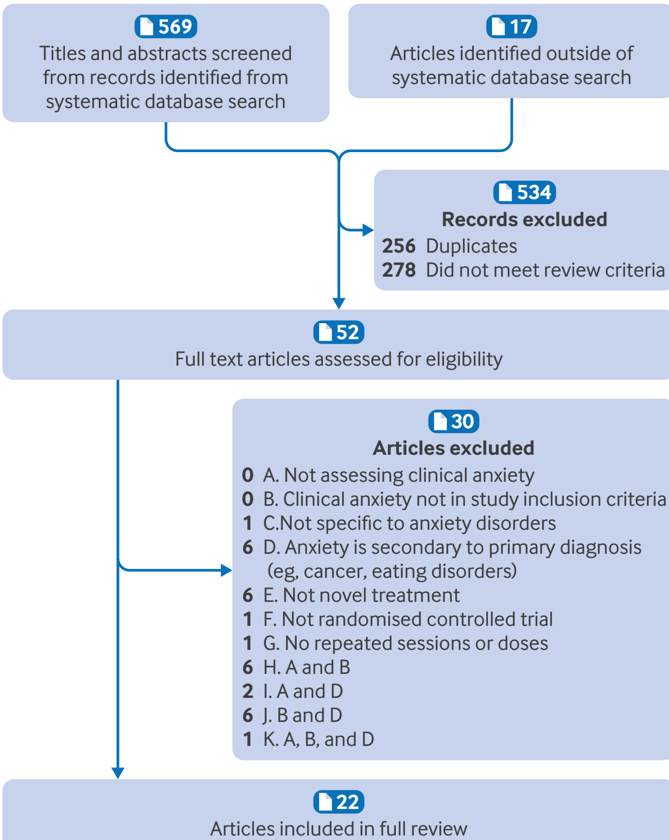

Fig 1 | Selection of articles for review of new and emerging treatments for anxiety disorders

anxiety disorder, generalized anxiety disorder, and panic disorder[17]), with effect sizes in comparison with placebo ranging from Cohen's d=0.23 to 0.60 for SSRIs and Cohen's d=0.50 to 0.53 for SNRIs.[17] Benzodiazepines are often prescribed given their rapid reduction in anxiety, with effect sizes in comparison with placebo ranging from Cohen's d=0.63 to 0.87, but their use incurs risks such as tachyphylaxis, misuse, physiological dependence, and overdose. Tricyclic antidepressants are among the earlier medications, tested primarily for panic disorder and generalized anxiety disorder. Although the tricyclics seem to be equally effective as SSRIs, with effect sizes compared with placebo at Cohen's d=0.74, they carry more side effects and risks, particularly in overdose. Off-label pharmacotherapies include certain anticonvulsants (such as pregabalin, Cohen's d=0.55, or gabapentin, limited data), antihistamines (such as hydroxyzine, Cohen's d=0.79), β-adrenergic blockade (eg, propranolol), and others (eg, mirtazapine, atypical antipsychotics, propranolol, and buspirone). In general, few studies directly compare pharmacotherapies for anxiety disorders.[17]

The most widely empirically supported psychotherapy is CBT, which is tailored to the specific features of each anxiety disorder and each person. Effect sizes are of a similar magnitude as pharmacotherapies. For example, of 69 randomized clinical trials (4118 outpatients), Hedges' *g* effect sizes for CBT compared with various control conditions (57 active and 12 waitlist) at post-treatment and at six and 12 months' follow-up were Hedge's *g*=0.07-0.40 for generalized anxiety disorder, *g*=0.22-0.35 for panic disorder with or without agoraphobia, *g*=0.34-0.60 for social anxiety disorder, and *g*=0.49-0.72 for specific phobia.[18] Effects were typically larger with waitlist control versus active comparisons. Fewer studies reported results later than 12 months' follow-up; the associations remained significant for generalized anxiety disorder (Hedge's *g*=0.22) and social anxiety disorder (Hedge's *g*=0.42) but not for panic disorder or agoraphobia and were not available for specific phobia. There is need for more high quality randomised controlled trials (RCTs) with more than 12 months of follow-up. Aside from CBT, interpersonal therapy,[19] panic-focused psychodynamic therapy, acceptance and commitment therapy (Hedge's *g*=0.52),[20] mindfulness-based stress reduction,[21] and relaxation techniques (Hedge's *g*=0.62) [22] have received at least some support, though often not as robust as for CBT (see supplemental materials for treatment descriptions for specific anxiety disorders). Additionally, cognitive bias modification yields small reductions in anxiety symptoms,[23] and exercise yields small to moderate reductions.[24 25]

Furthermore, some RCTs have examined combined pharmacological and psychotherapeutic treatment versus monotreatment—the results for which are mixed and heterogeneous. One meta-analysis found that there was no superiority of combined treatment in social anxiety disorder or generalized anxiety disorder, but it reported evidence that combined treatment could produce worse outcomes with panic disorder.[26] Another meta-analysis found that medications overall (including SSRIs, SNRIs, and tricyclic antidepressants) and combined pharmacological-psychotherapeutic treatment outperformed psychotherapies (including CBT, psychodynamic, therapies without face-to-face contact (eg, internet based), eye movement desensitization reprocessing, and interpersonal therapy).[27]

Not only are many with anxiety disorders undiagnosed,[28] but only about 25% of people diagnosed with an anxiety disorder receive treatment globally.[29] This sizeable treatment gap is partly due to the paucity of trained professionals.[30] Projections indicate that shortages of mental health practitioners will continue throughout the next decade, underscoring the need for innovative and scalable solutions to deliver evidence based therapies.[31] This shortage, along with other barriers to receiving care—such as financial strain, limited time, and the stigma around mental illness—have driven the burgeoning area of digital mental health. Internet therapies are now established as effective tools for managing anxiety, with the most evidence for CBT digital therapies,[21] and with little difference in outcomes compared with face-to-face CBT.[32 33] Digital therapies for anxiety are also widely accepted.[34] There is some

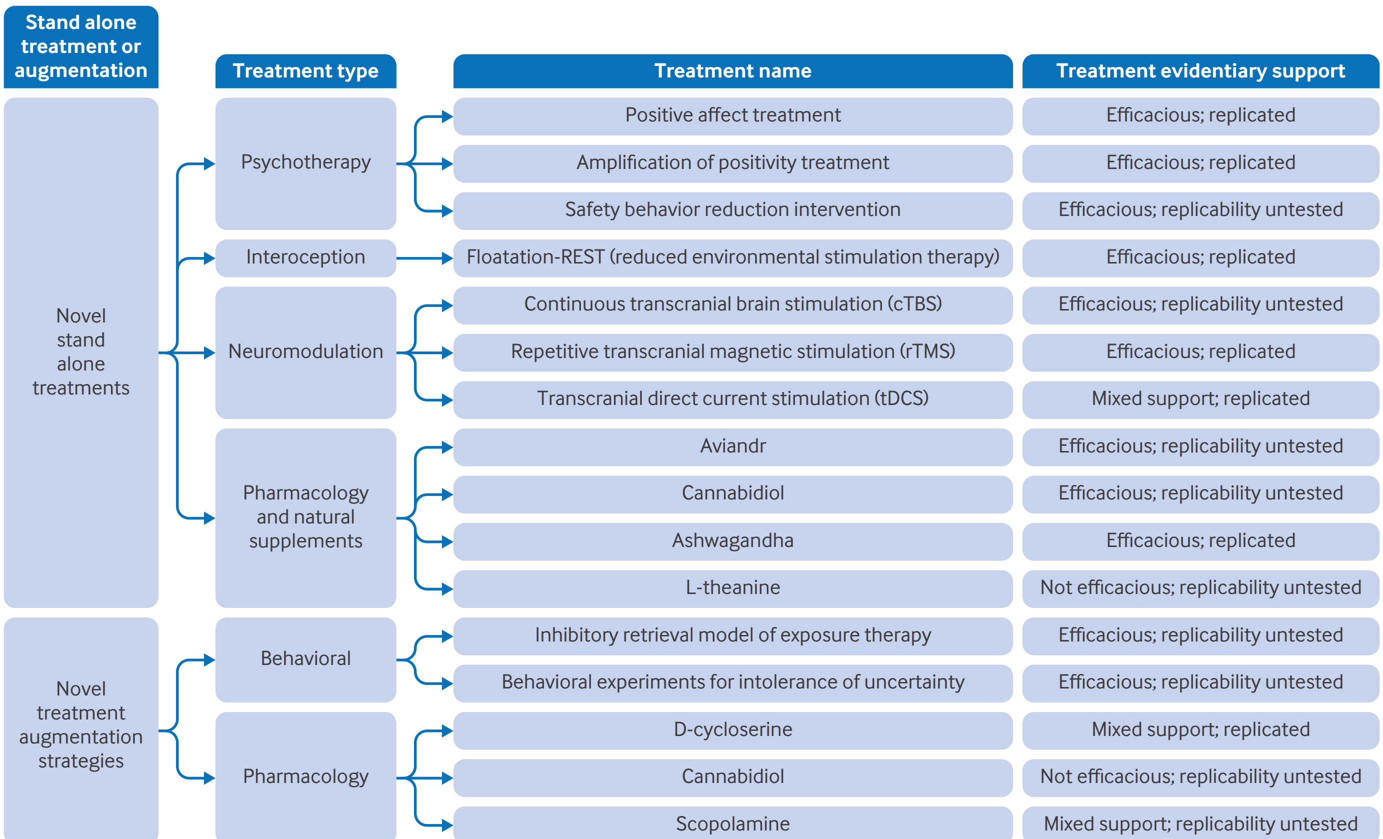

Fig 2 | Novel stand-alone treatments and augmentation strategies for anxiety disorders published between 2019 and 2024 Our chosen articles identified stand-alone treatments and treatment augmentation strategies. Novel stand-alone treatments included psychotherapies differing from traditional cognitive behavioral therapy, novel uses of neuromodulation, an interoception intervention, pharmacological agents, and natural supplements. All treatment augmentation strategies attempted to increase the efficacy of exposure therapy via pharmacological intervention or updated behavioral strategies. See supplemental tables 1 and 2.

evidence for larger effects when digital interventions are provided with support (via coaches, therapists, or administrators) relative to unguided interventions for anxiety, especially with synchronous versus asynchronous and more versus less support.[35 36] "Chatbot therapy," based on artificial intelligence, shows small reductions in anxiety symptoms (Hedge's g=0.19) with short term use, but it is understudied for empirical applications of CBT or long term use.[37] Additionally, clinical trials typically standardize the duration, quantity, or dose of treatment to examine efficacy. However, real world clinical practice outside of clinical trials often involves treatments that last months or years. This extended duration highlights the need to explore new treatment options that may be able to improve clinical effectiveness in real world settings in a shorter duration.

Novel stand-alone treatments

Reward based psychotherapy

Positive affect treatment (PAT) and amplification of positivity treatment (AMP)

Anhedonia (diminished interest or joy in usual activities) and underlying dysregulation of reward processing is characteristic of anxiety and depression,[38] and yet extant anxiety treatments typically fail to target reward mechanisms or are inadequate for improving positive affect. For example, a systematic review and meta-analysis found that conventional treatments (eg, CBT) have a much smaller effect size in increasing positive emotions (Hedge's g=0.27) than decreasing negative emotions (Hedge's g=−0.90).[39] Both PAT and AMP target cognitive, behavioral, and neurobiological reward processes that are important for increasing positive emotionality.[40] PAT aims to increase (i) anticipation of and response to reward, (ii) reward learning through a set of tailored exercises, including imaginal recounting and in-depth savoring of positive experiences attained through behavioral activation, (iii) attending to positive features and imagining positive outcomes of situations and attributing positive outcomes to self, and (iv) mindfulness based strategies to cultivate positivity, such as gratitude, generosity, and loving-kindness (fig 3).[41] AMP also aims to improve reward processing.[42] Specific AMP strategies include increasing awareness, exposure, and responsiveness to positive events (for example, savoring, reminiscing, and disclosing positive experiences to others); promoting the experience and expression of gratitude; and engaging in prosocial and kind acts toward others.

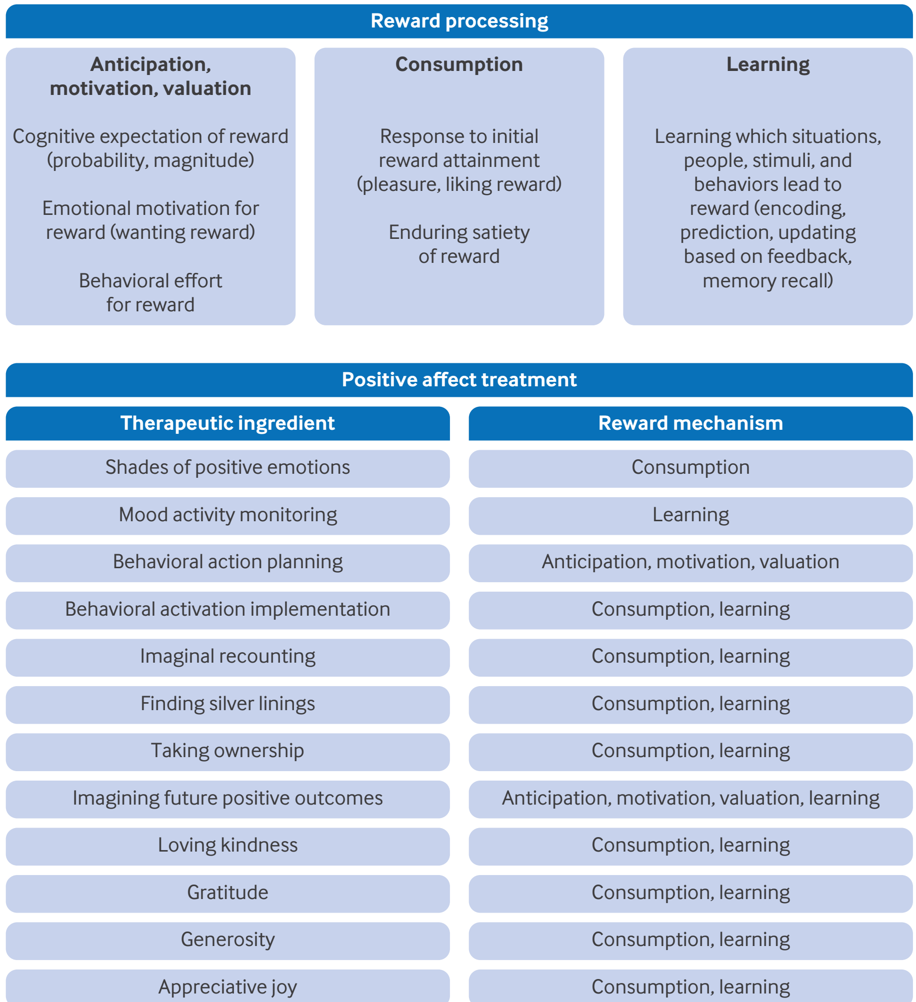

Fig 3 | Positive affect treatment: (a) components of reward processing and (b) examples of therapy

Effects

In the first RCT, with a sample of individuals who were anxious, depressed, or both, PAT significantly reduced anxiety, depression, stress (Depression, Anxiety, and Stress Scale – 21-Item Version (DASS-21)), and negative emotions to a greater degree at six months' follow-up than a CBT condition designed to decrease negative emotionality.[41] PAT also resulted in significantly greater increases in positive emotions than the comparison condition at post-treatment and six month follow-up. This comparison condition partly differed from standard CBT by exclusively focusing on reducing negative thoughts and emotions (but not attempting to increase positive thoughts and emotions). In a replication RCT that also evaluated reward processing mechanisms, PAT significantly improved positive emotions, stress, and depression (DASS-21) in comparison with the same CBT condition, though the two treatments improved anxiety and negative emotions to a similar degree. PAT also improved measures of reward processing, which were correlated with improvements in clinical outcomes.[43]

In the first RCT comparison with waitlist control,[42] AMP showed greater improvements in anxiety (composite score of the Overall Anxiety Severity and Impairment Scale (OASIS) and the State-Trait Anxiety Inventory–Trait Version (STAI-T)), positive emotions, depression, negative emotions, and psychological wellbeing. In a follow-up comparison with waitlist control,[44] AMP again showed greater improvements in anxiety (OASIS and PROMIS Anxiety), positive emotions, depression, negative emotions, and psychological wellbeing.

Overall, both PAT and AMP reflect a new, similar approach to improve reward processing, demonstrating promising methods for improving anhedonia and positive emotionality while reducing anxiety, depression, and negative emotionality more generally.[45]

Adverse effects

PAT resulted in no adverse or serious adverse events in either RCT.[41 43] In the first RCT, of the 41 participants randomized to PAT, 26 (63%) completed all 15 sessions. This was similar to the CBT comparison condition, to which 55 participants were randomized, and 31 (56%) completed all 15 sessions. In their second RCT,[43] attrition rates improved; 78.6% completed PAT, and 60.5% completed the CBT comparison condition. Similarly, AMP resulted in no adverse events in either RCT.[42 44] In its first RCT, 16 participants were randomized to AMP, and 15 (94%) completed treatment. In its second RCT, 37 participants started AMP, and 29 (78%) completed it.[44]

Clinical application

AMP has been tested mostly for social anxiety, whereas PAT targets anhedonia, anxiety, and depression more broadly. PAT and AMP have been administered largely by clinical psychology graduate students under the supervision of licensed clinical psychologists, across in-person or videoconferencing settings. PAT is designed to be 15 sessions,[41 43] whereas AMP was designed to be 10 sessions in its original version[42] or five or 10 sessions in its later form targeting social reward.[44] Given evidence from replications, clinical implementation is feasible.

Safety behavior reduction

"Safety behavior reduction"[46] is consistent with the inhibitory retrieval model of exposure therapy,[47 48] grounded in Pavlovian learning, which highlights the importance of reducing safety signals (ie, behaviors or contexts that lower the individual's expectation of their feared outcome). A key feature is that eliminating safety signals promotes prediction-error learning—a core mechanism of exposure therapy[49]—enabling patients to learn that their expected feared outcomes do not occur as predicted, resulting in anxiety reduction.

Effects

Only one published RCT has compared safety behavior reduction with a control intervention (text messages reminding individuals to focus on the present moment ("present-centered control")).[46] Safety behavior reduction involved a mostly automated text message intervention for socially anxious individuals to remind them to prevent use of social safety behaviors. Individuals identified their three most frequent safety behaviors from a larger list; then they received a total of 16 brief text message reminders at 12 pm to not engage in safety behaviors spread over every other day, for one month. This intervention did not provide suggestions to approach the social situations (as exposure therapy would), but rather to just refrain from safety behaviors. While effects between conditions were marginally significant ($P<0.10$) in reducing social anxiety at post-treatment, safety behavior reduction resulted in significant reduction in social anxiety at one month's follow-up.

Adverse effects

The presence or absence of adverse events was not reported in the RCT.[46] Of the 48 participants randomized to safety behavior reduction, 35 (73%) completed the treatment. This was similar to the present-centered control condition, to which 46 participants were randomized and 34 (74%) completed the treatment.

Clinical application

This intervention is automated and can be delivered to individuals with a cell phone and internet connection.[46] While it was tested as a stand-alone intervention, it could be used as a supplement to concurrent exposure therapy. However, replication is needed before clinical implementation.

Interoception

Floatation-REST (reduced environmental stimulation therapy)

Floatation-REST (also known as "float therapy") is a treatment that bears similarity to exposure therapy for internal physical sensations (ie, interoceptive sensations such as heart rate). During this intervention, individuals float on their backs in a shallow pool of water saturated with magnesium sulfate (Epsom salt) at 35°C (95°F). The floatation-REST environment minimizes sensory experience via attenuation of visual, auditory, olfactory, gustatory, tactile, thermal, vestibular, gravitational, and proprioceptive inputs, with reduced need for movement or speech. This results simultaneously in heightened awareness of interoceptive sensations and reduced levels of anxiety and stress.[50] Healthy individuals often report increased relaxation from before to after the floatation session.[51 52] Recent research has begun examining the effects of floatation-REST in individuals with heightened anxiety sensitivity, which refers to the fear of anxiety-related physical symptoms (eg, rapid heart rate, shortness of breath) due to beliefs that these symptoms may lead to harmful social, psychological, or physical outcomes.[53] Anxiety sensitivity is elevated in anxiety disorders—particularly panic disorder, post-traumatic stress disorder (PTSD),[53 54] agoraphobia, and generalized anxiety disorder (GAD)[54]—and is more strongly associated with these disorders than mood disorders.[54]

Effects

In an RCT of individuals with GAD,[55] participants completed 12 45-minute sessions over seven weeks (two sessions per week) compared with waitlist control. Floatation-REST resulted in greater improvement than the waitlist condition across multiple measures (self- reported GAD symptomatology, worry, emotion regulation, mindfulness, sleep, and depression). Another RCT with anxious or depressed individuals who conducted floatation-REST with flexible duration and frequency (compared with 1-hour weekly floatation-REST or zero-gravity chair REST) found that, across six sessions, all three REST conditions reduced anxiety measured at the beginning of sessions 5 and 6 (controlling for session 1 anxiety).[56] In this study, 97.3% of participants were diagnosed with major depressive disorder, and 50.7% were diagnosed with GAD. The study above with exclusively GAD

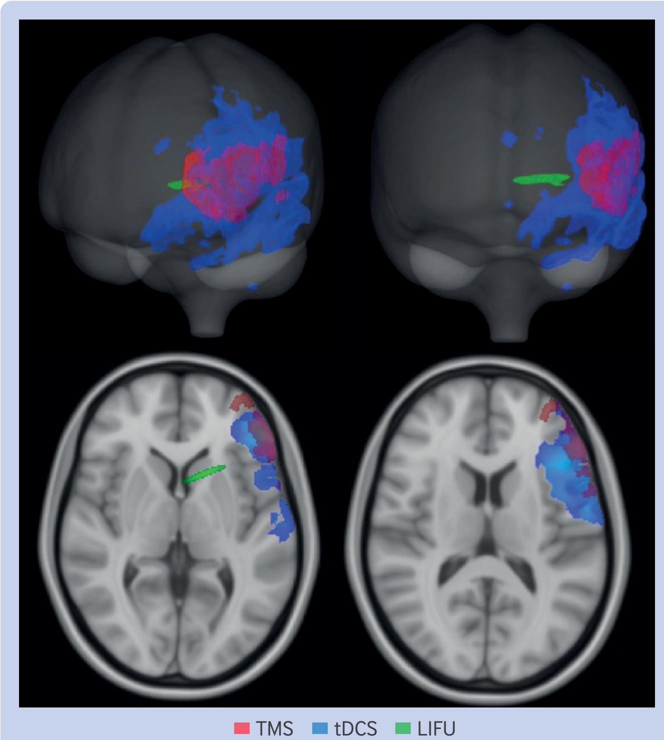

Fig 4 | Illustrative example of stimulation specificity and depth using modeling output of non-invasive neuromodulation techniques (TMS = transcranial magnetic stimulation; tDCS = transcranial direct current stimulation; LIFU = low intensity focused ultrasound). (Image from Kuhn T, Indahlastari A, Vila-Rodriguez F, et al[59]; adapted with permission). For each neuromodulation approach, highlighted regions represent areas with stimulation intensity above half-maximum

participants[55] found significant long term effects of floatation-REST compared with control, whereas the latter study[56] found similar long term anxiolytic effects between floatation-REST and a zero-gravity chair REST stringent control in a sample of 50.7% of individuals with GAD. This suggests that floatation-REST is likely helpful in individuals with GAD but that additional studies are needed.

Adverse effects

The safety and feasibility of a six-session floatation-REST treatment in anxious or depressed individuals showed that adherence was high[57]—with 89% adherence in individuals randomly assigned to floatation-REST with flexible frequency and duration (up to 90 minutes), and 85% in individuals with weekly 1-hour sessions; these were descriptively greater than adherence to a zero-gravity chair REST control condition (74%). Both floatation-REST conditions (1-hour weekly: 24%; flexible: 16%) had lower dropout than chair-REST (32%), though this was not significantly different. There were no serious adverse events reported related to any condition. Other studies similarly report no adverse events and that the floatation experience was pleasant.[50]

Clinical application

Floatation-REST has been dosed as a single session[58] or multi-session intervention (eg, six or 12 sessions),[55 57] with multi-session interventions often used to evaluate long term effects on anxiety. Prior studies have primarily focused on feasibility and have occurred at specialized facilities. There are few RCTs with primary outcomes focusing on efficacy (eg, anxiety reduction), but the emerging data suggest that further study is warranted. Replication of efficacy is needed before clinical implementation.

Neuromodulation

Neuromodulation techniques offer a promising approach for treating anxiety disorders by directly influencing brain activity, potentially helping to regulate the neural circuits involved in anxiety. There are several non-invasive neuromodulation techniques that vary according to depth of brain regions capable of being targeted (see fig 4 and table 1).[63] Repetitive transcranial magnetic stimulation (rTMS) and transcranial direct current stimulation (tDCS) have been used with anxiety disorders[60] but cannot reach deeper subcortical brain regions and provide diffuse stimulation to the brain.[66-68] Conversely, low intensity focused ultrasound (LIFU) is a novel approach capable of targeting surface level and deep subcortical brain regions with exceptionally high spatial precision[69 70] (see section on emerging treatments).

Continuous theta burst stimulation (cTBS), repetitive transcranial magnetic stimulation (rTMS), and transcranial direct current stimulation (tDCS)

The most common brain region targeted with these forms of neuromodulation is the right dorsolateral

Table 1 | Comparison of non-invasive neuromodulation techniques for anxiety disorders

Technique and protocol	Type of stimulation possible	Depth of stimulation possible	Target accuracy
Transcranial magnetic stimulation (TMS):			
Repetitive (rTMS)[60 61]	Excitatory Inhibitory	Shallow (cortex)	Diffuse
Continuous theta burst stimulation (cTBS)[61 62]	Inhibitory	Shallow (cortex)	Diffuse
Intermittent theta burst stimulation (iTBS)[60 61]	Excitatory	Shallow (cortex)	Diffuse
Transcranial direct current stimulation (tDCS):			
High frequency[60 63]	Excitatory	Shallow (cortex)	Diffuse
Low frequency[60 63]	Inhibitory	Shallow (cortex)	Diffuse
Low intensity focused ultrasound (LIFU)[59 64]	Excitatory Inhibitory	Deep	Specific

LIFU stimulation is inhibitory rather than excitatory for affective disorders,[65] including anxiety.[64]

prefrontal cortex (dlPFC).[62 71 72] The dlPFC is thought to be central to the cognitive control network,[73] and dlPFC-amygdala circuitry is involved in threat processing[71 74] with altered functioning in individuals with anxiety.[75 76] Neuromodulation of the dlPFC can directly affect dlPFC activity as well as downstream amygdala activity.[72]

Repetitive transcranial magnetic stimulation (rTMS) uses a magnetic field to repeatedly deliver electric currents to the brain, which can be used therapeutically and applied daily for up to six weeks.[63 77-80] High frequency rTMS (some define this as >5 Hz (cycles per second)[63] or >10 Hz[62]) causes cortical excitation, whereas low frequency (≤1 Hz) causes cortical inhibition.[81 82] rTMS has been used most often to treat major depressive disorder (approved by the US Food and Drug Administration (FDA) in 2008) and for obsessive-compulsive disorder (approved by the FDA in 2018),[79 83] but it has been used additionally with PTSD and GAD.[83-85] Theta burst stimulation (TBS) is an innovative and specific form of rTMS administered intermittently (iTBS) or continuously (cTBS). iTBS is delivered in bursts of three pulses at ultra-high frequency (50 Hz) with an inter-burst interval of 200 ms or 5 Hz (ie, theta frequency) and can increase cortical excitability and induce long term potentiation (ie, increase in connectivity between synapses, resulting in long lasting change in neural responding),[62 86] whereas cTBS is delivered continuously and can reduce cortical excitability and induce long term depression (ie, reduction in neural responding).[62 86] TBS has shorter administration times compared with other rTMS protocols (30× shorter),[60 61 87-89] increasing its efficiency and feasibility. For visualization of various TMS protocols, see Njenga et al.[90]

tDCS is the predominant protocol of transcranial electrical stimulation.[62] It involves the application of a mild electric current to the brain surface through electrodes on the scalp. It alters cortical excitability by modifying the resting potential of neural membranes that endure beyond the tDCS session. tDCS can increase (anodal stimulation) or decrease (cathodal stimulation) neural excitability.[91]

Effects

In a systematic review and meta-analysis of RCTs of non-invasive neuromodulation for GAD, cTBS and rTMS outperformed tDCS (tDCS did not demonstrate efficacy).[62] An RCT of 120 individuals with GAD was the only study to examine cTBS in comparison with 1 Hz rTMS and sham cTBS over the right dlPFC.[89] As assessed using the HAM-A, both cTBS and rTMS were superior to sham in reducing anxiety; cTBS also had lower anxiety at post-treatment and one month follow-up compared with rTMS. Additionally, there were significantly more responders at post-treatment and significantly more responders and remitters at one month follow-up with cTBS than rTMS, and, while both produced more responders than sham at post-treatment and follow-up, only cTBS had more remitters than sham at post-treatment (both cTBS and rTMS had more remitters than sham at follow-up). The results suggest cTBS and 1 Hz rTMS outperform sham and that cTBS is superior to 1 Hz rTMS for GAD, but replication is needed.

Not included in the review and meta-analysis[62] was a study that examined 1 and 2 mA tDCS of the left (anodal; excitatory) and medial (cathodal; inhibitory) PFC compared with sham in individuals with social anxiety disorder.[92] Results showed that both 1 mA and 2 mA tDCS conditions similarly reduced anxiety based on the Liebowitz Social Anxiety Scale (LSAS) compared with sham, but the 2 mA condition also improved worry, depression, and attentional bias to threat to a greater degree than the 1 mA condition. Overall, cTBS and rTMS seem to be effective for anxiety, whereas tDCS has mixed results. Given the shorter duration of cTBS and its efficacy thus far, replication and extension (eg, to other anxiety disorders) is warranted for this potentially impactful treatment.

Adverse effects

The most common adverse effects of TBS, rTMS, and tDCS in individuals with anxiety were headaches and dizziness; however, in comparison with sham control, only headaches were more prevalent in the active condition.[62] Active and sham participants did not significantly differ in their rates of discontinuing neuromodulation. One tDCS study found no major side effects but that pain, burning, and itching were experienced more in tDCS than sham.[93] Another tDCS study found no difference between 1 mA, 2 mA, and sham tDCS for any adverse effects.[92] One study comparing cTBS with rTMS reported both treatments had similar side effects that were greater than sham, including transient headaches and dizziness; no other side effects or adverse events occurred.[89] Thus, these neuromodulation techniques seem to mostly produce temporary, mild discomfort in some patients.

Clinical application

While neuromodulation is FDA-approved and used clinically for depression or OCD,[61] its use in anxiety disorders remains in the research stage and requires further evaluation before being implemented clinically.

Pharmacology and natural supplements

Maritupirdine (Aviandr)

Maritupirdine is a novel noradrenergic and specific serotonergic antidepressant evaluated in individuals with GAD at two doses (40 mg and 60 mg) compared with placebo.[94] Its mechanism of action differs from that of selective serotonin reuptake inhibitors (SSRIs). Maritupirdine is a potent antagonist of adrenergic α 2A, 2B, and 2C receptors, as well as serotonergic 5-HT2A, 5HT-2C, 5-HT6, and 5-HT7 receptors. Additionally, it functions as a histamine H1/H2 receptor antagonist without inverse agonist activity.[95]

Effects

Compared with placebo, both doses of maritupirdine improved anxiety (HAM-A) and clinician-rated improvement and symptom severity (Clinical Global Impression Scale) in individuals with GAD.[94]

Adverse effects

Due to its molecular composition, maritupirdine does not share the same adverse effects as an SSRI.[94] During the RCT, 34.1% of participants from all conditions reported adverse events. Participants in the placebo and low dose maritupirdine conditions reported similar frequency of adverse events, but the high dose maritupirdine participants reported more side effects than both of the other conditions. All side effects were mild to moderate; no severe adverse events were reported. The investigators considered the following side effects to be related to maritupirdine: dysgeusia (distortion of taste), upper abdominal pain, anxiety disorder, anxiety, dizziness, nausea, and somnolence. One participant permanently discontinued high dose maritupirdine due to experiencing five side effects; one participant permanently discontinued placebo due to experiencing four side effects. No adverse events affecting clinically important vital signs (blood pressure, sitting heart rate, or respiratory rate) were reported.

Clinical application

Maritupirdine is a new medication that is still in the research stage and is thus not clinically available. Further studies for replication, dosage, and safety are needed.

Ashwagandha

Withania somnifera (commonly known as ashwagandha) is an herb native to South Asia with a millennia-long history of use in Ayurvedic medicine for multiple ailments, including anxiety.[96 97] We include ashwagandha in this review because it has only recently gained attention in Western medicine through testing in RCTs for anxiety disorders.[98] Ashwagandha is thought to have anti-inflammatory and antioxidant properties and is believed to enhance gamma-aminobutyric acid (GABA) activity—the primary inhibitory neurotransmitter involved in anxiety regulation, including inhibitory effects in the amygdala.[98-104]

Effects

A recent systematic review and meta-analysis[103] found two studies examining ashwagandha versus placebo in clinically anxious individuals,[103 105] though we were unable to locate one of the clinical trials (cited as Khyati and Anup, 2013). In the available report,[105] there was a significantly greater clinical response for participants receiving ashwagandha (88.2%) compared with placebo (50%) at week 6 of treatment, with “clinical response” defined a priori as a HAM-A score of ≤12 and a Global Rating Scale score from both participant and rater of ≤1, indicating mild symptoms. Scores solely from the HAM-A trended toward significance in favor of the ashwagandha condition. In another RCT[100] individuals with GAD who were all receiving SSRIs were randomized to additionally receive either ashwagandha or placebo. The ashwagandha condition resulted in significantly greater reductions in anxiety (HAM-A) compared with placebo.

Adverse effects

A recent systematic review and meta-analysis examining the safety and efficacy of Ashwagandha [106] found no differences in side effects between Ashwagandha and placebo in one study,[105] with the most common side effects including drowsiness and heaviness of head (i.e., grogginess). Another study reported no adverse effects of Ashwagandha.[100] A third study similarly reported no adverse events when used in individuals with insomnia.[107]

Clinical application

The doses of Ashwagandha used in the clinical studies reviewed vary widely, including 600 mg/day (primarily assessing sleep and with anxiety a secondary outcome),[107] 1000 mg/day,[100] or adjusting dose based on patient response.[105] Due to the limited number of RCTs, an optimal therapeutic dose remains unclear. Future research could clarify the optimal dose of ashwagandha, as well as its efficacy as a monotreatment. Ashwagandha is available over the counter, allowing individuals to access it independently. However, treatment is commonly overseen by Ayurvedic or naturopathic medicine practitioners.

L-theanine

L-theanine is a non-protein amino acid found in *Camellia sinensis* tea leaves.[108 109] Preclinical studies suggest that it may reduce excitatory glutamate activity while enhancing GABA, glycine, dopamine, and α wave brain activity, which are associated with anxiolytic effects.[110-113]

Effects

One RCT [108] examined the effects of 8-week L-theanine versus placebo as an adjunctive treatment in individuals with GAD undergoing stable antidepressant treatment. L-theanine did not significantly reduce anxiety compared with placebo as measured by the HAM-A.

Adverse effects

The one clinical trial targeting anxiety showed no significant adverse events for L-theanine versus placebo.[108] One meta-analysis examining L-theanine across psychiatric disorders more broadly similarly found no significant adverse events.[113 114]

Clinical application

L-theanine has been evaluated in only one RCT for anxiety, which showed no significant efficacy. Thus, there is currently no rationale for clinical implementation. However, future studies could explore its efficacy as a monotreatment.

Novel treatment augmentation strategies

The only novel treatment augmentation strategies that have been published are behavioral and pharmacological approaches for improving exposure therapy. Most target Pavlovian extinction learning principles, from which exposure therapy was derived (see box 3).

Behavioral augmentation of exposure therapy

Inhibitory retrieval model of exposure therapy

The inhibitory retrieval model of exposure therapy is the most recent theoretical framework for exposure therapy.[47-49] It posits that exposure therapy primarily operates through Pavlovian prediction-error learning, leading to extinction, and that various processes then guide retrieval of that extinction memory across time and contexts (ie, inhibitory learning and retrieval).[47-49] In other words, the model emphasizes learning that feared outcomes do not occur as expected across a variety of situations. Evidence supporting the purported mechanisms is accruing.

In a large sample of individuals with social anxiety disorder, panic disorder, agoraphobia, or specific phobias, expectancy change—the degree to which individuals reduce their expectations about feared outcomes following exposure—was predictive of anxiety symptom reduction.[116] Similarly, a higher proportion of exposures in which the feared outcome did not occur was predictive of anxiety reduction.[117] An RCT comparing the inhibitory retrieval model with a habituation based model of exposure therapy found that changes in expectations that the feared outcome would occur during exposures (but not changes in fear during exposures) predicted anxiety symptom reduction at post-treatment and three month follow-up in the inhibitory retrieval condition.[118]

Unlike fear habituation models of exposure therapy, the inhibitory retrieval model incorporates strategies designed to maximize expectancy violation (ie, prediction error) and retrieval of extinction memories (ie, memories that the anxiety-provoking situation is safe, not dangerous) in new contexts. These include removing safety signals, maintaining attention to the target stimulus during the exposure (eg, focusing on the audience during a public speech exposure), rehearsing inhibitory learning, varying stimuli and contexts, using deepened extinction, positive occasion setter extinction, incorporating occasional reinforced extinction, and employing mental reinstatement sparingly.[47 48]

Effects

Although aspects of the inhibitory retrieval model have been evaluated across several studies, only one RCT directly compared the full inhibitory retrieval model with another form of exposure therapy (exposure therapy focused on habituation/fear reduction).[119] In a sample of individuals with social anxiety disorder or panic disorder, both conditions resulted in anxiety symptom reduction, but the inhibitory retrieval condition resulted in greater reductions of anxiety and lower anxiety at post-treatment based on anxiety questionnaires, and greater reductions in anxiety during the behavioral approach task (an in vivo public speech task).

Adverse effects

The presence or absence of adverse events was not reported.[119] Of the 47 participants randomized to inhibitory retrieval exposure therapy, 36 (77%) completed the nine-session treatment. For comparison, of the 42 participants randomized to habituation exposure therapy, 29 (69%) completed the nine-session treatment.

Clinical application

Inhibitory retrieval exposure therapy has been tested with social anxiety and panic disorder but awaits testing with other anxiety disorders. It has been delivered primarily by clinical psychology graduate students under the supervision of licensed clinical psychologists, across in-person or videoconferencing settings. The treatment has been tested in a nine-session format but can be extended based on patient needs. Given the strong empirical support for exposure therapy for anxiety disorders[120] and evidence suggesting that inhibitory retrieval exposure therapy demonstrates moderately superior efficacy compared with habituation-focused exposure therapy,[119] clinical implementation is feasible.

Behavioral experiments for intolerance of uncertainty

The behavioral experiments for intolerance of uncertainty intervention[121] aligns closely with

Box 3: Pavlovian extinction and exposure therapy

Pavlovian learning offers strong theoretical bases for explaining the development and treatment of anxiety disorders.[47-49 115] Pavlovian acquisition occurs when an otherwise neutral stimulus is repeatedly paired with a biologically relevant unconditional stimulus (US), causing the neutral stimulus to become a conditional stimulus (CS) that predicts the occurrence of the US. When the US is an aversive stimulus, individuals will form conditional fear and anxiety (conditional response; CR) towards the CS in anticipation of the US that will follow. For example, individuals with public speaking anxiety may fear (CR) that giving a speech (CS) will result in social rejection (US). Pavlovian extinction occurs when the CS is repeatedly presented without the US, which results in a reduction of the CR towards the CS. For example, repeatedly giving public speeches (CS) without social rejection (US) can reduce public speaking anxiety (CR). Exposure therapy is the clinical analogue of Pavlovian extinction. The latest model of exposure therapy—the inhibitory retrieval model[48]—suggests that exposure therapy operates by learning that the CS no longer predicts the US to the degree that was initially expected (ie, prediction error) and that this learning is gated by the context in which it occurs. Through repeated exposure in varied contexts, the anxious individual reduces their expectation that the CS will lead to the US and therefore reduces their CR.

exposure therapy based on the inhibitory retrieval model.[47 48] Behavioral experiments for intolerance of uncertainty is conceptually related to exposure therapy but differ in their primary target and learning mechanisms. Whereas exposure based on inhibitory learning theory aims to disconfirm threat expectancies about specific feared outcomes (eg, injury, rejection), intolerance of uncertainty exercises focus on building tolerance for uncertainty itself by placing individuals in ambiguous, anxiety-provoking situations and discouraging reassurance-seeking. Success is measured by expectancy violation or increased willingness to experience uncertainty across a broader set of outcome predictions than with the inhibitory retrieval model (eg, "What will happen?" "How will I feel?"), making these exercises particularly suited for GAD, where pervasive intolerance of uncertainty is a core factor. Thus, both are forms of exposure therapy, but they stem from partially different conceptual models.

Behavioral experiments for intolerance of uncertainty involves 12 weekly one-hour sessions in which participants conduct behavioral experiments (ie, exposures) and gradually reduce safety behaviors to test whether their catastrophic predictions about certain situations materialize and whether their emotional responses unfold as expected. This differs from the safety behavior reduction treatment described in the stand-alone treatments because that one involves only safety behavior reduction (no exposures), whereas behavioral experiments for intolerance of uncertainty involves exposures and safety behavior reduction.

Effects

An RCT evaluated behavioral experiments for intolerance of uncertainty in individuals with GAD. Compared with waitlist control, the behavioral experiments for intolerance uncertainty resulted in improved anxiety (clinical severity rating from the anxiety disorders interview schedule for the DSM-IV) from pre-treatment to post-treatment, which endured through the 12 month follow-up assessment.[121] Secondary measures were questionnaires of anxiety, worry, intolerance of uncertainty, and depression—all of which similarly saw greater improvement in the treatment condition compared to waitlist control from pre- to post-treatment and through 12 month follow-up.

Adverse effects

The presence or absence of adverse events was not reported.[121] Of the 30 participants randomized to behavioral experiments for intolerance of uncertainty, 23 (77%) completed the treatment.

Clinical application

Behavioral experiments for intolerance of uncertainty was designed for and tested in individuals with GAD and administered by clinical psychology graduate students under the supervision of licensed clinical psychologists.[121] It is designed to be a 12-session treatment. Replication is needed before clinical implementation is feasible.

Pharmacological augmentation of therapy

The pharmacological agents reviewed here aim to enhance the efficacy of exposure therapy. While their hypothesized mechanisms of action vary, they are generally intended to facilitate extinction learning, thereby improving exposure treatment outcomes (ie, anxiety reduction). Several pharmacological agents were not included in this review either because they have not been tested in recent clinical trials of anxiety disorders (eg, L-3,4-dihydroxyphenylalanine (L-DOPA), 3,4-methylenedioxy methamphetamine (MDMA), yohimbine, ketamine) or because they are not considered novel (eg, glucocorticoids). We included two exceptions of treatments that are not novel: *D*-cycloserine and cannabidiol. We included *D*-cycloserine because research is ongoing regarding its efficacy, and we included cannabidiol because it has recently undergone novel testing as an exposure therapy augmentation strategy, and patients providing feedback on this review showed interest in cannabis.

D-cycloserine

N-methyl-D-asparate (NMDA) receptor activity plays a crucial role in extinction learning.[122] NMDA antagonists impair extinction learning, whereas NMDA agonists enhance it.[123] *D*-cycloserine is a partial NMDA agonist that enhances NMDA receptor function when glycine levels are low, promoting synaptic plasticity and potentially facilitating extinction learning and memory consolidation.[124 125] Improved extinction learning (eg, during exposure therapy) could reduce anxiety symptomatology.

Effects

A meta-analysis found that *D*-cycloserine moderately improved exposure therapy.[126] However, a more recent systematic review and meta-analysis of anxiety disorders, OCD, and PTSD reported a small and more nuanced effect at post-treatment, with no significant benefits at follow-up.[127] Subsequent studies have yielded similar findings, showing little to no enhancement of exposure therapy compared with control conditions.[128 129] Thus, *D*-cycloserine's effects seem inconsistent, often modest, or not significantly different from control. An important caveat is that all *D*-cycloserine studies to date used fear habituation models of exposure therapy. It has not yet have been tested with the inhibitory retrieval model, which is more closely aligned with extinction learning principles and may be more aligned with *D*-cycloserine.

Adverse effects

D-cycloserine is generally well tolerated, with no significant adverse effects reported in one review and meta-analysis.[130] Commonly reported side effects include headaches, drowsiness, confusion, tremors, memory difficulties, paresthesias, seizure, diarrhea, constipation, tiredness, dizziness, vomiting, and pain.[130 131] However, at low doses, these side effects are rare.[130] A recent RCT found that *D*-cycloserine was not associated with a higher incidence of side effects compared with placebo. No serious adverse events were reported, and participants were generally satisfied with the treatment.[131] Thus, *D*-cycloserine seems to be well tolerated with limited side effects.

Clinical application

A recent review of international guidelines found that there are no established guidelines for the use of *D*-cycloserine in clinical practice.[132] This reflects its inconsistent efficacy, experimental status in therapy, and insufficient supporting evidence to warrant clinical recommendations.

Scopolamine

Scopolamine is a muscarinic cholinergic receptor antagonist that has demonstrated greater effects on contextual rather than cued fear in adult rats, likely due to cholinergic blockade in the hippocampus.[133-135] In rodents, low doses selectively affect contextual learning, whereas higher doses affect learning more generally.[133 136] Preclinical research suggests that scopolamine reduces the renewal of fear after extinction, meaning that fear is less likely to return when the anxiety-provoking stimulus is encountered in a different context than where the extinction of the stimulus occurred.[137] This fear-reducing effect is thought to result from hippocampal down regulation when scopolamine is administered shortly before extinction, weakening the processing of the extinction context. Since extinction learning is typically context-dependent, scopolamine may help make extinction learning transfer across contexts, increasing generalization across different environments and thereby reducing anxiety symptomatology and relapse after exposure therapy.

Effects

One RCT[138] evaluated the effects of 0.5 or 0.6 mg scopolamine versus placebo in socially anxious individuals with public speaking anxiety undergoing VR-based exposure therapy (seven sessions of seven exposures each). Compared with placebo, both doses of scopolamine lowered skin conductance (a measure of physiological anxious arousal) at exposure onset and termination, as well as during anticipation and recovery phases. However, the two doses did not differ from each other in their effects on skin conductance. Additionally, scopolamine did not significant differ from placebo on self reported anxiety measures.

Adverse effects

The placebo group reported less intense scopolamine-related side effects and less distress than both scopolamine groups, with no significant differences between the two scopolamine groups. No adverse or serious adverse events were reported by participants in the RCT. Overall, side effects appeared to be mild in both intensity and distress.

Clinical application

There has only been one human RCT evaluating the potential efficacy of scopolamine in anxiety. As a result, its potential clinical use remains experimental and awaits further research.

Cannabidiol

Cannabidiol is the only treatment in our review that was evaluated both as a stand-alone treatment and a treatment augmentation strategy. For concision, we only include information on cannabidiol in the treatment augmentation section of this review.

Cannabidiol is a component of the *Cannabis sativa* plan (commonly known as cannabis, marijuana, or hemp) that does not produce psychoactive symptoms.[139] Cannabidiol acts on the human endocannabinoid system[140]—particularly cannabinoid receptors type 1 and type 2[141]—which is thought to regulate anxiety.[142] In Pavlovian fear conditioning experiments, the endocannabinoid system has been shown to be involved in extinction learning with rodents[143] and humans,[144 145] which is relevant for exposure therapy.

Effects

One recent systematic review found three RCTs of cannabidiol versus placebo in individuals with social anxiety disorder, but two of these studies only assessed short term anxiety during tasks (eg, during a public speaking task, other anxiety-provoking tasks).[146] One RCT of teenagers with social anxiety disorder and comorbid avoidant personality disorder received cannabidiol (300 mg) or placebo for four weeks. The cannabidiol condition significantly reduced social anxiety as measured by the Fear of Negative Evaluation Questionnaire and the Liebowitz Social Anxiety Scale.[147] Another RCT evaluated whether cannabidiol enhanced eight sessions of exposure therapy compared with placebo in individuals with social anxiety disorder or panic disorder with agoraphobia.[148] Results showed that there were no effects of cannabidiol compared with placebo through the six month follow-up, though both conditions reduced anxiety—likely due to exposure therapy.

Adverse effects

Based on a systematic review in healthy individuals or individuals with clinical anxiety or sleep difficulty, cannabidiol is generally well tolerated without major adverse effects; the most common being fatigue and sedation.[149] The RCT examining cannabidiol effects on exposure therapy found no serious adverse events, and 10 adverse events were judged by blinded researchers and therapists to be "not related" or "possibly related" to the study medication.[148] These adverse events were not significantly greater in the cannabidiol or placebo conditions (cannabidiol n=4, placebo n=6). Lastly, the RCT evaluating cannabidiol versus placebo on social anxiety disorder did not systematically evaluate side effects, though they state that none of the participants reported any significant health complaints.[147]

Clinical application

Dosage ranges widely, between 25 and 900 mg.[146 149] Short term anxiolytic effects occur along an inverted U-shaped dosing curve (300 mg *v* 100 mg and 900 mg),[150] though the RCT assessing cannabidiol and placebo adjunct to exposure therapy found that 300 mg did not affect long term anxiety reduction.[148] Further research is needed to determine recommended dose and to determine whether cannabidiol is effective as a stand-alone treatment or as an exposure therapy augmentation strategy.

Emerging treatments

We identified three treatments that did not meet the review criteria but warrant mentioning: low intensity focused ultrasound (LIFU), psilocybin, and augmented reality (AR) for exposure therapy. LIFU is a neuromodulation technique that was found in an open-label clinical trial to reduce treatment-resistant GAD, as measured by the HAM-A and Beck Anxiety Inventory.[64] Second, psilocybin (commonly known as “magic mushrooms”) has been shown in three studies to reduce illness related anxiety,[151] but there are no studies to date for principal anxiety disorders. Lastly, augmented reality—which mixes virtually created objects with reality—has been shown in single-session interventions to reduce spider phobia[152 153] and social anxiety.[154] See supplementary material for details.

Guidelines

Currently, neither the US Department of Veterans Affairs nor the American Psychological Association provide clinical practice guidelines for anxiety disorders. The American Psychiatric Association (APxA) provides guidelines only for panic disorder,[155] recommending SSRIs, SNRIs, TCAs, and CBT. Among the novel stand-alone or augmentation strategies reviewed herein, the APxA guidelines mention *D*-cycloserine as an area of future research. For pediatric populations, the American Academy of Child and Adolescent Psychiatry[156] recommends CBT and SSRIs or SNRIs for panic disorder, social anxiety disorder, GAD, and separation anxiety disorder in individuals aged 6-18 years, without mention any of the novel strategies reviewed here.

Overall, practice guidelines for anxiety disorders have remained largely unchanged over the past 10-15 years. The most comprehensive practice guidelines for anxiety disorders are provided by the National Institute for Health and Care Excellence (NICE). The NICE guidelines for GAD and panic disorder (last updated in 2020)[157] and social anxiety disorder and specific phobia (published in 2013)[158] do not reference any of the novel stand-alone or augmentation strategies discussed in this review. In all cases, treatment recommendations center on CBT and SSRIs or SNRIs. The British Association of Psychopharmacology guidelines[159] mention *D*-cycloserine as a potential method for augmenting exposure for panic disorder, social anxiety, and specific phobias.

Conclusion

The number of novel treatments and treatment augmentation strategies for anxiety disorders is encouraging. All treatments reviewed herein except L-theanine demonstrate at least some efficacy in reducing anxiety. The strongest evidence exists for reward-based psychotherapies—positive affect treatment (PAT) and amplification of positivity treatment (AMP)—as these are separate treatments, both independently developed based on reward principles by two laboratories, and each laboratory replicated their results in reducing anxiety.[41-44] Other treatments demonstrating efficacy in at least two laboratories include floatation-REST and ashwagandha. Several treatments showed efficacy in a single RCT, including safety behavior reduction, cTBS, maritupirdine (Aviandr), behavioral experiments for intolerance of uncertainty, scopolamine, and the inhibitory retrieval model of exposure therapy. *D*-cycloserine has mixed effect sizes across many RCTs. Lastly, cannabidiol was tested in two RCTs as a stand-alone treatment (efficacious) and an augmentation strategy for exposure therapy (not efficacious).

The most innovative non-pharmacologic approaches include reward-based psychotherapies and floatation-REST due to their novel therapeutic perspectives and varying techniques. Maritupirdine is the only novel medication we found targeting anxiety.

The diversity of treatments is also encouraging. Psychotherapy and psychopharmacology are represented in novel formats. New and emerging neuromodulation strategies such as LIFU and cTBS introduce an entirely different treatment paradigm. Floatation-REST provides a therapist-free approach to interoceptive exposure.

This review will hopefully encourage further research into these novel treatments. Many of these treatments require replication and extension across

QUESTIONS FOR FUTURE RESEARCH

- Which treatments reviewed here show robust efficacy through replication and extension across anxiety and related disorders?
- If neuromodulation treatments consistently demonstrate efficacy, what are the barriers to their clinical implementation (eg, training, equipment acquisition, cost), and how can they be integrated into standard care?
- With an expanding range of treatments available for anxiety, can we develop predictive models or biomarkers to optimize first-line treatment selection for individual patients?

HOW PATIENTS WERE INVOLVED IN THE CREATION OF THIS ARTICLE

We received feedback from individuals with anxiety to ensure the review accurately reflects patients' perspectives on emerging treatments. Several patients highlighted the potential of positive affect treatment (PAT) and amplification of positivity treatment (AMP), noting that, while the concepts seem simple, structured professional guidance could make them more impactful. Others expressed curiosity about floatation-REST, with some perceiving it as highly relaxing, while others had reservations due to personal discomfort with water or enclosed spaces. Neuromodulation techniques were seen as promising but difficult to fully grasp, while the inclusion of non-SSRI pharmacological options, such as maritupirdine, was welcomed. L-theanine was of particular interest, though some found its placement in the article unclear. Patients also discussed the complex relationship between cannabis and anxiety, noting that responses varied widely, with some experiencing heightened anxiety. Many appreciated the comprehensive nature of the review but expressed a desire for more concrete examples of how certain interventions are applied in practice. One patient summarized their perspective on treatment innovation: "It was a breath of fresh air to hear about a medication in clinical trials for anxiety that is NOT considered an SSRI." Another emphasized the importance of shifting treatment goals, stating, "I think increasing positive emotions, rather than merely decreasing negative emotions, is an important goal in the treatment of anxiety and depression."

anxiety disorders. Perhaps the interventions that consistently demonstrate efficacy and effectiveness may enter standard practice, and further research could address treatment personalization—which patient would respond best to which treatment, including how to treat affective dimensions (eg, depression, anhedonia) that commonly co-occur with anxiety disorders. Additionally, future directions for research may involve artificial intelligence (eg, clinical decision making about which treatment to provide and when to provide treatment).

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- TDZ, SSK, and MGC all made substantial contributions to the contents of the work. TDZ conducted the literature review, wrote the majority of the first draft, and created all figures (with acknowledged contributions for fig 4), tables, and most boxes. SSK contributed treatments for potential inclusion in the first draft and acquired, synthesized, and wrote the box for patient feedback of the manuscript. MGC wrote much of the introductory sections and guideline section in the first draft.
- TDZ, SSK, and MGC all critically reviewed and edited the manuscript in its entirety for important intellectual content.
- TDZ, SSK, and MGC all approved the manuscript.
- TDZ, SSK, and MGC agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Supplemental material: Disorder-specific treatment details

Supplemental tables 1 and 2: Details of studies included in review of novel stand-alone treatments and augmentation strategies for anxiety disorders